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Simulation of a Pneumatic Pressure Measurement System Using MATLAB Simulink

LAB PROJECT REPORT

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Aircraft Systems and Instrumentation
Course Code: AS254IA

Department of Aerospace Engineering
RV College of Engineering®

December 21, 2025

RV COLLEGE OF ENGINEERING®, BENGALURU-59

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DEPARTMENT OF AEROSPACE ENGINEERING

CERTIFICATE

Certified that the lab project titled **"Simulation of a Pneumatic Pressure Measurement System Using MATLAB Simulink"** has been carried out by **Akula Uday Kiran (1RV23AS004) Kiran V Airani (1RV23AS020) Tejas L (1RV23AS061) Vishnu Raghavendra Mannapur (1RV23AS063)** students of the Department of Aerospace Engineering, **R V College of Engineering, Bengaluru**, who are bonafide students of **Visvesvaraya Technological University, Belagavi**, in partial fulfilment of the requirements for the award of the degree of **Bachelor of Engineering** in **Aerospace Engineering** during the academic year **2025–2026**.

It is further certified that all corrections and suggestions indicated for the internal assessment have been incorporated in the lab project report deposited in the department. The project work has been approved as it satisfies the academic requirements prescribed by the institution for the said degree.

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DECLARATION

We, the students of V Semester, Department of Aerospace Engineering, R V College of Engineering, Bengaluru–560059, hereby declare that the lab project titled **"Simulation of a Pneumatic Pressure Measurement System Using MATLAB Simulink"** has been carried out by us and submitted in partial fulfilment of the requirements for the award of the Degree of **Bachelor of Engineering** in **Aerospace Engineering** of the Visvesvaraya Technological University, Belagavi during the academic year **2025–2026**.

We further declare that the content of this lab project report has not been submitted previously by anybody for the award of any Degree or Diploma to any other University or Institution.

We also declare that any intellectual property rights generated out of this project carried out at RV College of Engineering will be the property of **R V College of Engineering, Bengaluru**, and we will be among the authors of the same.

Place: Bengaluru

Date:

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ABSTRACT

The advancement of green propulsion systems, particularly hydrogen peroxide (H_2O_2)-based monopropellants, is essential for ensuring environmentally sustainable and safer space missions. However, the performance and reliability of such systems are contingent upon robust ground testing methodologies. This project focuses on the design, development, and validation of a modular propulsion system test bench tailored for expulsion testing under controlled laboratory conditions. The bench integrates critical subsystems such as a calibrated load cell, pressure reducing orifice, mechanical support structures, and interface assemblies—all organized through a systems engineering approach.

A central feature of the setup is the pulley-based load cell calibration mechanism. A clamp-mounted pulley system was designed to apply known weights to the load cell, enabling precise force application and allowing the generation of an accurate force–voltage calibration curve. This mechanical system was structurally analyzed and fabricated to ensure safety, stability, and repeatability. In parallel, a load cell support structure was developed for proper alignment and secure integration within the test bench, ensuring minimal deformation during thrust events.

To manage internal fluid dynamics, a pressure reducing orifice was designed and implemented. Using flow rate equations and iterative simulations, the orifice dimensions were selected to provide a specific pressure drop, ensuring safe and consistent operation. Following analysis, the component was procured and integrated into the fluid line. To replicate real propulsion conditions, a detailed expulsion test plan was developed. This plan defined operational constraints, selected key test variables (e.g., pulse width, chamber pressure, thrust), and outlined multiple test scenarios. A data acquisition system was deployed to record sensor outputs and thrust readings in real time. A critical element throughout the project was interface design, ensuring robust mechanical and sensor connectivity between all subsystems. These interfaces were modeled, analyzed, and physically validated for precision alignment and durability. The system integration was guided by a detailed Piping and Instrumentation Diagram (P&ID), which facilitated smooth assembly, troubleshooting, and scalability. This work culminates in a versatile, reliable test bench for evaluating H_2O_2-based propulsion units and serves as a model framework for academic and early-stage propulsion testing.

ACKNOWLEDGEMENT

We would like to express our sincere gratitude to our project guide **Gp.Capt. Deepak Bana VSM (RETD)**, **Visiting Professor**, Department of Aerospace Engineering, R V College of Engineering, Bengaluru, for the valuable guidance, constant encouragement, and technical support provided throughout the course of this lab project.

We would like to thank **Dr. Supreeth R**, Associate Professor and Head, Department of Aerospace Engineering, R V College of Engineering, for providing the necessary facilities and a supportive academic environment to successfully complete this work.

We also express our sincere thanks to **Dr. K. N. Subramanya**, Principal, R V College of Engineering, Bengaluru, for his encouragement and support toward academic and laboratory activities.

We extend our gratitude to all the faculty members and technical staff of the Department of Aerospace Engineering for their assistance and cooperation during the laboratory sessions.

Finally, we would like to thank our parents and friends for their constant support and encouragement throughout the completion of this project.

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CHAPTER 1

INTRODUCTION

1.1 Introduction

Pneumatic systems are fundamental in modern industrial automation, manufacturing, and laboratory environments due to the widespread availability of compressed air and its inherent advantages such as nonflammability, cost-effectiveness, and ease of storage and conditioning. As a result, pneumatic actuation and control are extensively used in applications ranging from automated assembly lines and packaging systems to process control, instrumentation, and experimental laboratory setups.

A complete pneumatic measurement and control system typically consists of several interconnected subsystems. These include an air compressor driven by an electric motor to generate compressed air, a storage tank (receiver) to accumulate and stabilize pressure, control and safety elements such as valves and relief mechanisms to ensure operation within permissible limits, and precision pressure sensors to measure system pressure. These sensors convert the physical pressure inside the system into standardized electrical signals that can be transmitted, monitored, displayed, and used for control purposes. The ability to accurately measure, display, and regulate pneumatic pressure in real time is critical for ensuring system safety, optimizing performance, and meeting operational and regulatory requirements in industries such as automotive manufacturing, aerospace, pharmaceuticals, chemical processing, and building automation.

With increasing system complexity and the demand for higher reliability, simulation-based modeling has become an essential step in pneumatic system design and analysis. Instead of relying solely on physical experimentation, engineers increasingly use simulation tools to predict system behavior, evaluate design choices, and identify potential issues before implementation. MATLAB Simulink provides a block-diagram-based environment that enables intuitive modeling of multidisciplinary systems, allowing electrical, mechanical, and pneumatic components to be represented in a form that closely mirrors real-world signal and energy flow.

This project focuses on the development of a MATLAB Simulink-based simulation model of a pneumatic pressure generation and monitoring system. The model captures the complete signal chain from electrical input voltage to pneumatic pressure generation and finally to industrial-standard pressure sensing and display. An electrically driven positive-displacement compressor supplies air to a storage tank, where pressure builds up from atmospheric conditions.

A safety relief mechanism limits maximum pressure to a predefined threshold of 10 bar, ensuring safe operation. The resulting pressure is measured using a pressure sensor model that converts pressure into a 4–20 mA current signal, which is then displayed using both analog and digital indicators.

The system is intentionally modeled as a simplified practical representation, avoiding unnecessary thermodynamic and mechanical complexity while preserving the essential physical behavior required for educational and laboratory analysis. This balance allows clear understanding of system operation while maintaining realism and industrial relevance.

1.2 Background and Motivation

In industrial pneumatic installations, pressure is rarely used directly as a raw physical quantity. Instead, it is measured, conditioned, transmitted, and interpreted through standardized electrical signals that interface with control systems such as programmable logic controllers (PLCs), distributed control systems (DCS), and supervisory monitoring software. Among these standards, the 4–20 mA current loop has become the dominant method for analog signal transmission in industrial instrumentation.

The widespread adoption of the 4–20 mA standard is due to several practical advantages. The loop current is directly proportional to the measured variable, making signal interpretation straightforward. Current-based signaling is highly immune to electrical noise, particularly over long cable runs, which is essential in harsh industrial environments. Additionally, the use of a non-zero lower limit (4 mA instead of 0 mA) allows immediate fault detection: a current below 4 mA typically indicates broken wiring, sensor failure, or power loss.

In a typical pressure transmitter, applied pressure causes a mechanical deformation of a sensing element, such as a diaphragm. This deformation alters an electrical property—commonly resistance or capacitance—which is then processed by internal electronics. The electronics amplify, linearize, and scale the signal so that the minimum calibrated pressure corresponds to 4 mA and the maximum calibrated pressure corresponds to 20 mA. This standardized output enables seamless integration with industrial control and monitoring systems and has led to widespread adoption across maritime systems, chemical processing plants, power generation facilities, and automation-driven buildings.

From an educational standpoint, however, students often encounter pressure sensors and current loops as abstract concepts rather than as integrated parts of a complete system. The motivation behind this project is to connect theory with system-level understanding by modeling the entire pneumatic measurement chain in a single simulation environment. By doing so, learners can observe how electrical inputs influence mechanical motion, how airflow translates into pressure accumulation, how safety mechanisms protect the system, and how physical pressure is ultimately converted into a standardized electrical signal suitable for display and control.

1.3 Role of Simulation Using Simulink

Simulink and Simscape, developed by MathWorks, are powerful block-diagram-based simulation environments that enable engineers and students to model complex dynamic systems in a unified graphical framework. These tools support the modeling of physical domains such as electrical systems, mechanical components, and pneumatic networks alongside signal-processing and control logic.

One of the key strengths of Simulink is its ability to represent real-world hardware components as modular blocks, allowing users to create custom blocksets that mirror actual system architecture. This approach supports rapid prototyping, design verification, and virtual testing, significantly reducing development time and cost. For educational and research-oriented projects, simulation models provide a visual and interactive means to understand system dynamics and causal relationships between variables such as voltage, speed, flow rate, pressure, and current.

In this project, custom Simulink blocks are developed to represent the motor–compressor system, the pneumatic storage tank, the safety relief mechanism, and the pressure sensor with 4–20 mA signal conditioning. This structured modeling approach allows step-by-step visualization of system operation, including motor speed control, compressor flow generation, pressure build-up, relief valve activation, sensor signal generation, and final display output. As a result, the simulation serves not only as a functional model but also as an educational tool that clearly demonstrates both physical principles and practical engineering implementation details.

1.4 Aim and Objectives

Aim

The aim of this project is to develop and analyze a MATLAB Simulink model of a pneumatic pressure generation and monitoring system that demonstrates the interaction between electrical input, compressor operation, air storage, safety mechanisms, and industrial-standard pressure sensing.

Objectives

- To model a simplified practical positive-displacement compressor whose speed is controlled by input voltage.
- To calculate volumetric air flow based on compressor geometry, speed, and efficiency.

- To simulate pressure build-up in a storage tank starting from atmospheric conditions.
- To incorporate a safety relief valve that activates when tank pressure exceeds 10 bar.
- To implement a pressure sensor model using 4–20 mA current scaling.
- To visualize system behavior using analog gauges, digital displays, and time-domain plots.

1.5 System Overview and Functional Description

The complete pneumatic system is organized into clearly defined functional blocks within Simulink. Electrical input voltage serves as the primary control variable. This voltage determines the rotational speed of the motor, which in turn drives the compressor. The compressor generates airflow that is directed into a storage tank, causing pressure to rise over time.

As pressure accumulates, the system continuously monitors tank pressure using a pressure sensor block. When the pressure reaches a predefined threshold of bar, a safety relief mechanism becomes active, preventing further pressure increase and ensuring safe operation. The measured pressure is converted into a corresponding current signal following the 4–20 mA standard and displayed using both analog and digital indicators.

This modular structure makes the system easy to understand, modify, and extend for further experimentation.

1.6 Workflow of the Project

The project was carried out in a structured manner:

1. Understanding the physical behavior of pneumatic compressors and tanks.
2. Developing a mathematical model for compressor flow rate.
3. Implementing tank pressure dynamics using a first-order differential equation.
4. Adding safety relief logic based on pressure threshold.
5. Designing pressure and current sensing blocks.
6. Integrating all subsystems into a single Simulink model.
7. Observing system response through simulations and plots.

1.7 Assumptions Made in the Model

To keep the model suitable for laboratory-level simulation, the following assumptions are made:

- Air behaves as an ideal gas.
- Compressor dynamics such as torque ripple and valve delay are neglected.
- Flow losses in piping are ignored except for safety relief action.
- Tank pressure is uniform throughout the volume.
- Sensor dynamics are ideal, with linear scaling.

These assumptions simplify the model while preserving the essential system behavior.

CHAPTER 2

SYSTEM DESIGN AND METHODOLOGY

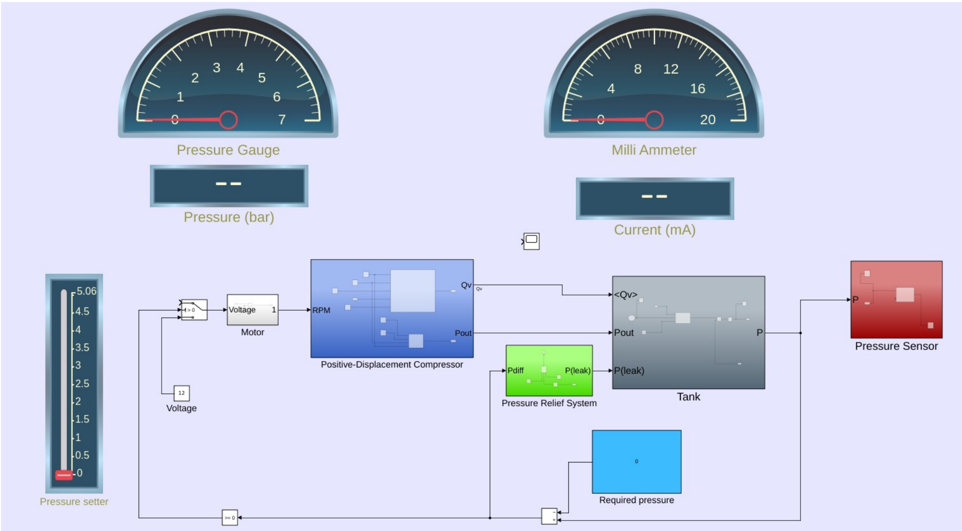

Figure 2.1: Pneumatic pressure measurement system overview

This project presents a Simulink-based model of a pressure control system using a positive displacement air compressor. The system is designed to generate, store, monitor, and regulate compressed air pressure inside a tank while ensuring safe operation through pressure relief and electrical monitoring.

The system operates according to a sequential, stagewise execution model that mimics real-world pneumatic system operation. As shown in the accompanying block diagram, the system can be conceptually divided into three primary subsystems: generation, storage and measurement, and signal conditioning and display.

2.1 Subsystem Overview

1. **Electrical Power Supply and Drive:**The process begins with an electric motor block that converts electrical input voltage (provided as a control signal) into mechanical rotational power. This motor drives a custom positive-displacement compressor block that draws in ambient air and compresses it to elevated pressure.

2. **Positive-Displacement Compressor:**The compressor block implements the fundamental relationship between motor speed (in revolutions per minute), compressor displacement (volume swept per revolution), and the volumetric flow rate of compressed air delivered.

3. **Tank(Air storage system:)**The tank serves as a storage vessel for compressed air. As the compressor supplies air, pressure within the tank increases according to the tank volume and airflow dynamics. The tank subsystem models pressure accumulation and transient behavior under varying compressor input and relief conditions.

4. **Pressure Relief System:**A pressure relief subsystem is implemented to ensure operational safety. When the system pressure exceeds a predefined threshold, the relief mechanism activates to vent excess air. This prevents damage to the storage tank and downstream components due to over-pressurization.

5. **Pressure Sensing and conversion to current:**A pressure sensor continuously measures the internal pressure of the tank. The sensor output represents the actual system pressure and is used both for monitoring and for signal conversion.
   The measured pressure is converted into a proportional electrical current signal using a signal conditioning block. This current represents the pressure value and follows a linear relationship, typically aligned with industrial standards 4–20 mA.

6. **Measurement and Visualization:**

   - A pressure gauge and digital display present the tank pressure in bar.
   - A milliammeter and digital display indicate the pressure-proportional current signal.
   - These visualization tools provide real-time insight into system behavior during simulation.

7. **Reference Pressure and Pressure setter:**A reference pressure value is included to define the required or target pressure. This signal is used for system evaluation, and future closed-loop control implementation.

2.1.1 Electrical Power Supply and Drive

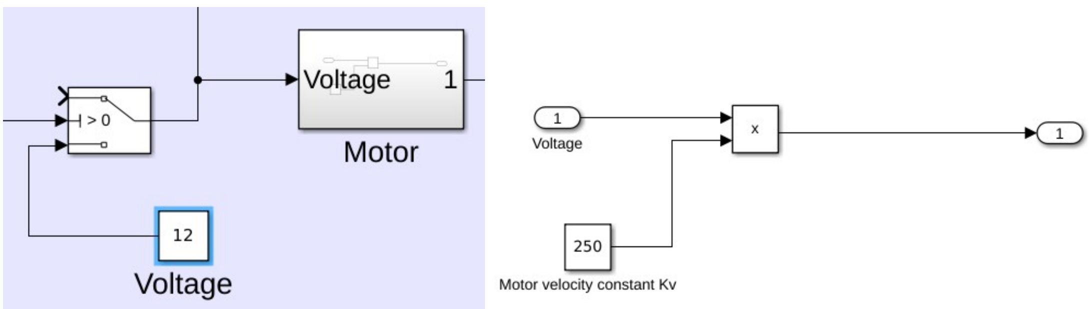

Figure 2.2: Power supply and motor drive assembly

Power Supply and Motor Drive Assembly

The power supply and motor drive assembly is responsible for providing controlled electrical input to the motor based on system operating conditions. A DC voltage source supplies a constant input voltage, which represents the available electrical power for driving the compressor. A switching block is incorporated in the drive path to enable or disable the voltage supplied to the motor. This switch is controlled by a feedback signal derived from the pressure sensing and comparison logic. When the measured pressure reaches or exceeds a predefined threshold, the switch opens, there-by cutting off the voltage supply to the motor and stopping compressor operation. When the pressure falls below a lower or newly defined threshold, the switch closes, re-energizing the motor and restarting the compressor.

This switching mechanism introduces an on–off (bang-bang) control behavior with hysteresis, which prevents frequent switching and ensures stable operation. It effectively regulates the pressure within the desired operating range without the need for complex control algorithms.

Motor Sub-Assembly

The motor sub-assembly models the dynamic relationship between the applied voltage and the motor's rotational speed. The motor is represented using a simplified linear model, where the motor speed is directly proportional to the applied voltage.

The relationship is defined as:

$$\omega = K_v \cdot V$$

where:

- ω motor angular velocity in RPM
- V DC voltage supplied
- K_v motor velocity constant

In the implemented model, the motor velocity constant K_v is set to 250 RPM/V, meaning that for every volt applied, the motor speed increases by 250 RPM. The multiplication block computes the motor speed by multiplying the input voltage with the velocity constant.

The output of the motor sub-assembly is the motor speed signal, which is supplied as the mechanical input to the positive displacement compressor. This abstraction neglects electrical transients and losses, allowing the focus to remain on pressure dynamics and system-level behavior.

Role of the Switch in the Feedback System

The switch plays a critical role in implementing the feedback-based pressure control strategy. Its operation is governed by the pressure feedback signal obtained from the pressure sensor and comparator logic. The control logic operates as follows:

- When tank pressure reaches the upper threshold, the switch opens and disconnects the motor from the voltage supply, stopping the compressor.

- When the required pressure is set higher than the previous threshold, or when pressure falls below the lower threshold, the switch closes, restoring voltage to the motor and restarting the compressor.

This approach ensures:

- Automatic pressure regulation
- Energy-efficient operation
- Protection against over-pressurization
- Reduced mechanical wear due to controlled switching

2.1.2 Positive-Displacement Compressor subsystem

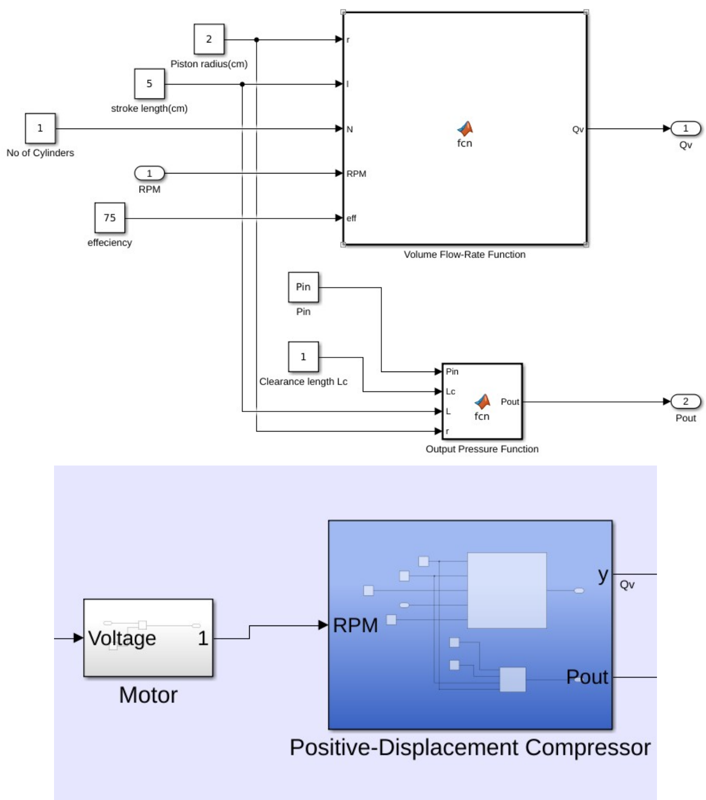

Figure 2.3: Positive Displacement compressor model

The positive displacement compressor subsystem models the pneumatic behavior of a multi-cylinder reciprocating compressor. It converts mechanical input in the form of motor rotational speed into compressed air flow and output pressure. The subsystem is implemented using parameterized MATLAB Function blocks to represent volumetric flow rate generation and pressure rise within the compression chamber.

Subsystem Inputs and Parameters

The compressor subsystem receives the following inputs and parameters:

Table 2.1: Motor and Positive Displacement Compressor Parameters

Subsystem	Parameter	Symbol	Value
Motor	Input voltage	V	12 V
Motor	Motor velocity constant	K_v	250 RPM/V
Motor	Output speed	RPM	$K_v \times V$
Compressor	Piston radius	r	2 cm
Compressor	Stroke length	L	5 cm
Compressor	Number of cylinders	N	1
Compressor	Volumetric efficiency	eff	75%
Compressor	Clearance length	L_c	1 cm
Compressor	Inlet pressure	P_{in}	1.013 bar

These parameters define the geometric and operational characteristics of the compressor.

Volume Flow Rate Computation

The volumetric flow rate of air delivered by the compressor is calculated using the Volume Flow-Rate Function implemented in the MATLAB Function block.

Mathematical Model

The piston cross-sectional area is given by:

$$A = 4\pi r^2 \tag{2.1}$$

The swept volume per stroke is calculated as:

$$V = A \cdot L \tag{2.2}$$

The volumetric flow rate is expressed as:

$$Q_v = \frac{V \cdot N \cdot RPM}{60} \cdot \frac{eff}{100} \cdot \frac{1}{1000} \tag{2.3}$$

where Q_v represents the volumetric flow rate in m^3/s. The factor 60 converts rotational speed from revolutions per minute to revolutions per second, while the scaling factor ensures unit consistency.

Output Pressure Computation

The compressor outlet pressure is computed based on the inlet pressure and volumetric compression within the cylinder.

Mathematical Model

The clearance volume is defined as:

$$V_c = \pi r^2 L_c \tag{2.4}$$

The swept volume is given by:

$$V = \pi r^2 L \tag{2.5}$$

The outlet pressure is calculated as:

$$P_{out} = P_{in} \cdot \frac{V}{V_c} \tag{2.6}$$

This relationship represents the pressure rise due to volumetric compression, assuming idealized compression behavior. **Subsystem Output Signals** The positive displacement compressor subsystem generates the following output signals:

- Volumetric flow rate, Q_v, supplied to the downstream tank
- Outlet pressure, P_{out}, representing the compressor discharge pressure

2.1.3 Tank(Air Storage System)

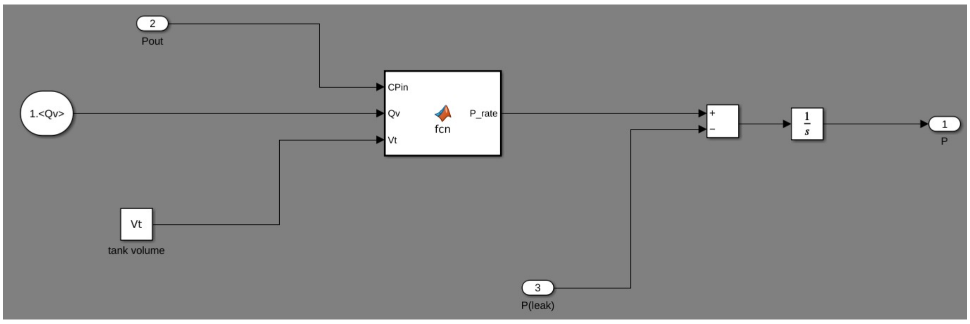

Figure 2.4: Air storage subsystem

The tank subsystem models the pressure accumulation dynamics of a compressed air storage vessel. It represents the relationship between incoming compressed air flow, pressure rise due to accumulation, and pressure reduction due to leakage or pressure relief mechanisms. The subsystem integrates the net pressure rate over time to obtain the instantaneous tank pressure.

Subsystem Inputs and Output

The tank subsystem consists of three input signals and one output signal.

Input Signals

- Compressor outlet pressure, P_{out}: Pressure input from the positive displacement compressor

- Volumetric flow rate, Q_v: Airflow rate supplied by the compressor to the tank
- Pressure leakage rate, P_{leak}: Rate of pressure drop due to the pressure relief system or leakage paths

Output Signal

- Tank pressure, P: Accumulated pressure within the tank over time

Pressure Rate Computation

The rate of pressure rise inside the tank is computed using a MATLAB Function block. The function is defined as:

$$P_{\text{rate}} = \frac{dP}{dt} = \frac{P_{in}}{V_t} \cdot Q_v \tag{2.7}$$

where:

- $P_{\text{rate}} = \frac{dP}{dt}$ is the rate of pressure increase
- P_{in} is the pressure input from the compressor
- Q_v is the volumetric flow rate
- V_t is the tank volume

Net Pressure Dynamics

The pressure rise rate is combined with the pressure leakage rate using a summation block. The leakage pressure signal is subtracted from the pressure rise rate to obtain the net pressure rate:

$$\frac{dP}{dt} = P_{\text{rate}} - P_{\text{leak}} \tag{2.8}$$

This formulation allows the model to account for both pressure accumulation and pressure loss mechanisms.

Pressure Integration

The net pressure rate is integrated over time using an integrator block ($1/s$) to obtain the instantaneous pressure within the tank:

$$P(t) = \int \left(\frac{dP}{dt}\right) dt \tag{2.9}$$

The integrator output represents the continuously varying pressure stored in the tank.

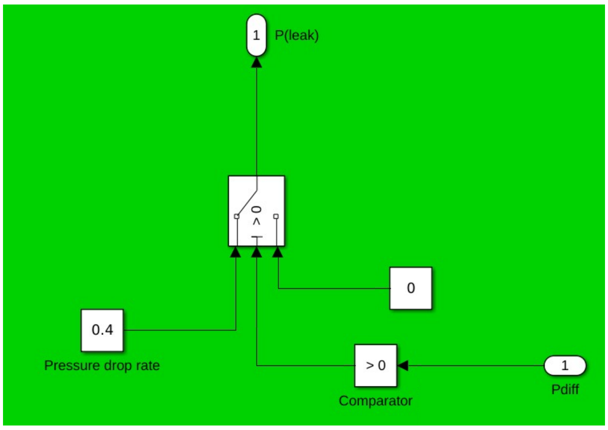

Figure 2.5: Pressure relief subystem

2.1.4 Pressure Relief System

The pressure relief subsystem is designed to ensure safe operation of the compressed air system by preventing excessive pressure buildup within the storage tank. It dynamically relieves pressure whenever the tank pressure exceeds the required operating limit. The subsystem operates based on a pressure difference signal derived from the comparison between the measured tank pressure and the predefined required pressure.

Subsystem Functionality

The primary function of the pressure relief subsystem is to generate a pressure leakage rate signal that represents controlled pressure discharge from the tank. This leakage signal is applied to the tank subsystem to reduce the accumulated pressure when necessary. Pressure relief is activated only during excess pressure conditions, thereby minimizing unnecessary pressure loss.

Pressure Difference Evaluation

The subsystem receives the pressure difference signal, P_{diff}, which is computed as the difference between the required pressure and the actual tank pressure measured by the pressure sensor.

$$P_{diff} = P_{req} - P \tag{2.10}$$

A comparator block evaluates the sign of this pressure difference and operates the switching logic:

- $P_{\text{diff}} < 0$: Tank pressure exceeds the required pressure, the switch outputs a predefined pressure drop rate, enabling pressure relief
- $P_{\text{diff}} \geq 0$: Tank pressure is within or below the allowable limit, the switch outputs zero, indicating no pressure relief action

This evaluation forms the basis of the pressure relief decision logic.

Pressure Drop Rate
A constant pressure drop rate is defined within the subsystem to represent the rate at which pressure is relieved from the tank during overpressure conditions. This value models the behavior of a pressure relief valve or controlled venting mechanism. **Subsystem Output**

Signal
The pressure relief subsystem generates a single output signal:

- Pressure leakage rate, P_{leak}: A non-negative signal representing the rate of pressure reduction applied to the tank

This signal is subtracted from the pressure accumulation rate within the tank subsystem to regulate the tank pressure.

2.1.5 Pressure sensing and signal processing

The pressure sensing and signal processing subsystem is responsible for measuring the pressure accumulated in the tank and converting it into a standardized electrical current signal suitable for monitoring, display, and interfacing with external systems. This subsystem represents the instrumentation and signal conditioning stage of the overall pressure control system.

Subsystem Description
The subsystem receives the tank pressure signal as its input. This pressure signal is displayed locally using a pressure gauge and simultaneously processed by a MATLAB Function block to generate a proportional current output. The generated current signal represents the measured pressure in an electrical form commonly used in industrial automation and control systems.

Pressure Measurement
The tank pressure output is connected to a pressure sensor block, which models an ideal pressure transducer. The sensor output corresponds directly to the measured pressure in bar. This signal is used for both visualization and further signal processing. A pressure gauge and a digital display provide real-time indication of the measured pressure.

Signal Processing and Pressure-to-Current Conversion
The signal processing block implements a linear pressure-to-current conversion using a MATLAB Function. The conversion follows a standardized industrial current transmission format, mapping the pressure range to a milliampere (mA) signal.

The implemented conversion relationship is expressed as:

$$I = I_{low} + \left(\frac{I_{high} - I_{low}}{P_{high} - P_{low}} \right) P \tag{2.11}$$

where:

- I is the output current in milliamperes (mA)
- P is the measured pressure in bar

This equation scales the pressure signal to a standard **4–20 mA** current range, where 4 mA corresponds to zero pressure and 20 mA corresponds to the maximum rated pressure. This approach ensures reliable signal transmission, improved noise immunity, and compatibility with industrial control and monitoring systems.

CHAPTER 3

IMPLEMENTATION AND RESULTS

3.1 Model Implementation in MATLAB Simulink

The complete pneumatic pressure control system described in the previous chapter was implemented in MATLAB Simulink under the model name *PneumaticsMain*. The model integrates electrical input control, motor dynamics, positive-displacement compression, pneumatic storage, safety relief logic, pressure sensing, signal conditioning, and visualization into a single executable simulation framework.
Each subsystem was first developed and verified independently to ensure correct operation. After subsystem-level verification, all components were interconnected following the defined signal flow. Particular attention was given to unit consistency, signal routing, and initialization of state variables, especially within the tank pressure integrator block.
The simulation was executed using a continuous-time solver with default step-size settings, which is suitable for capturing gradual pressure dynamics and steady-state behavior in pneumatic systems.

3.2 Initial Conditions and Parameter Setup

Before executing the simulation, the following initial conditions and system parameters were defined:

- Initial tank pressure: Atmospheric pressure
- Supply voltage: 6 V
- Motor velocity constant: 250 RPM/V
- Pressure relief threshold: 10 bar
- Pressure sensor output range: 4–20 mA

These values were selected to reflect realistic laboratory-scale pneumatic operation while ensuring safe system behavior and clear observation of pressure dynamics.

3.3 Simulation Sequence of Operation

At the start of the simulation, the electrical voltage source supplies input to the motor through the switching logic. The motor converts the applied voltage into rotational speed, which drives the positive-displacement compressor. The compressor generates compressed air flow proportional to motor speed and delivers it to the storage tank.
As air accumulates in the tank, internal pressure rises gradually from atmospheric conditions. The tank pressure is continuously monitored by the pressure sensor subsystem. When the pressure approaches or exceeds the predefined safety threshold, the comparison logic and pressure relief system are activated. This results in either disconnection of the motor supply or controlled venting of excess pressure, thereby preventing further pressure increase. Throughout the simulation, pressure and current signals are displayed using gauge indicators, digital readouts, and scope blocks for time-domain observation.

3.4 Results and Observations

The simulation results demonstrate stable and predictable system behavior consistent with pneumatic system theory.

Figure 3.1: Scope for Compressor Flow Rate Dynamics (blue line)

3.4.1 Tank Pressure Build-Up Relief Valve Action

The tank pressure increases smoothly from atmospheric pressure once the compressor is energized. The rate of pressure rise depends on the motor speed and corresponding compressor airflow. When the safety threshold of 10 bar is reached, pressure stabilization is observed due to activation of the relief system. The pressure relief mechanism effectively limits the maximum tank pressure. The relief action prevents further pressure increase beyond the safety limit and ensures safe operation of the system.

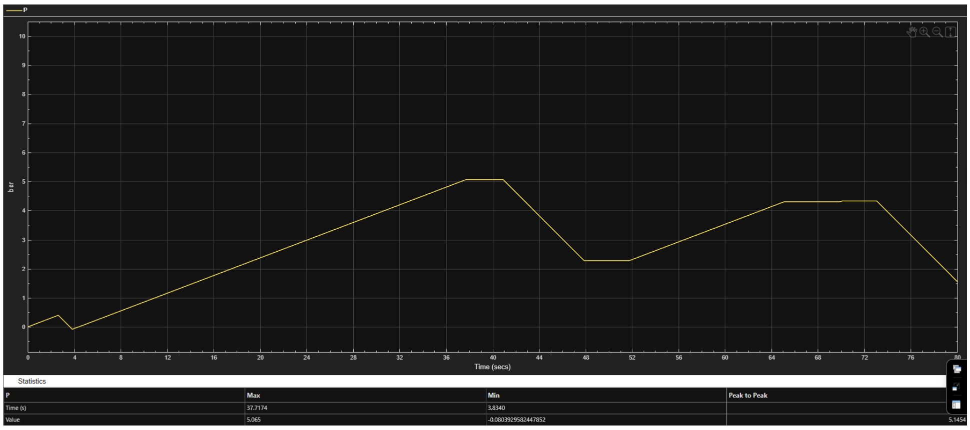

Figure 3.2: Pressure Response Showing Relief Valve Activation

3.4.2 Pressure Sensor Current Output

The pressure sensor output exhibits a linear relationship between tank pressure and current output. As pressure increases, the sensor current rises proportionally from 4 mA toward 20 mA, confirming correct implementation of the 4–20 mA signal conditioning.

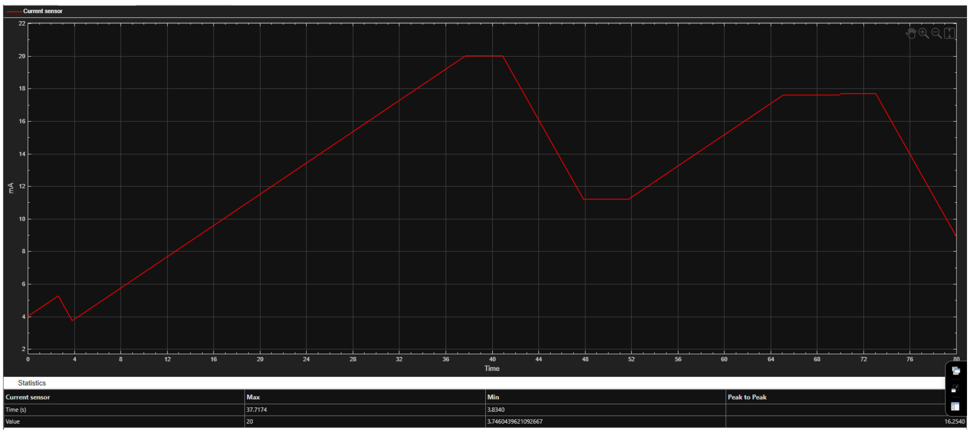

Figure 3.3: Pressure-to-Current (4–20 mA) Conversion Characteristics

3.5 Discussion

The observed results validate the modeling approach adopted in this project. The simplified motor and compressor representations generate realistic pressure dynamics while maintaining computational efficiency. The on–off control logic and pressure relief system provide effective pressure regulation without instability or oscillation. The pressure sensor subsystem accurately converts physical pressure into an industry-standard electrical signal.

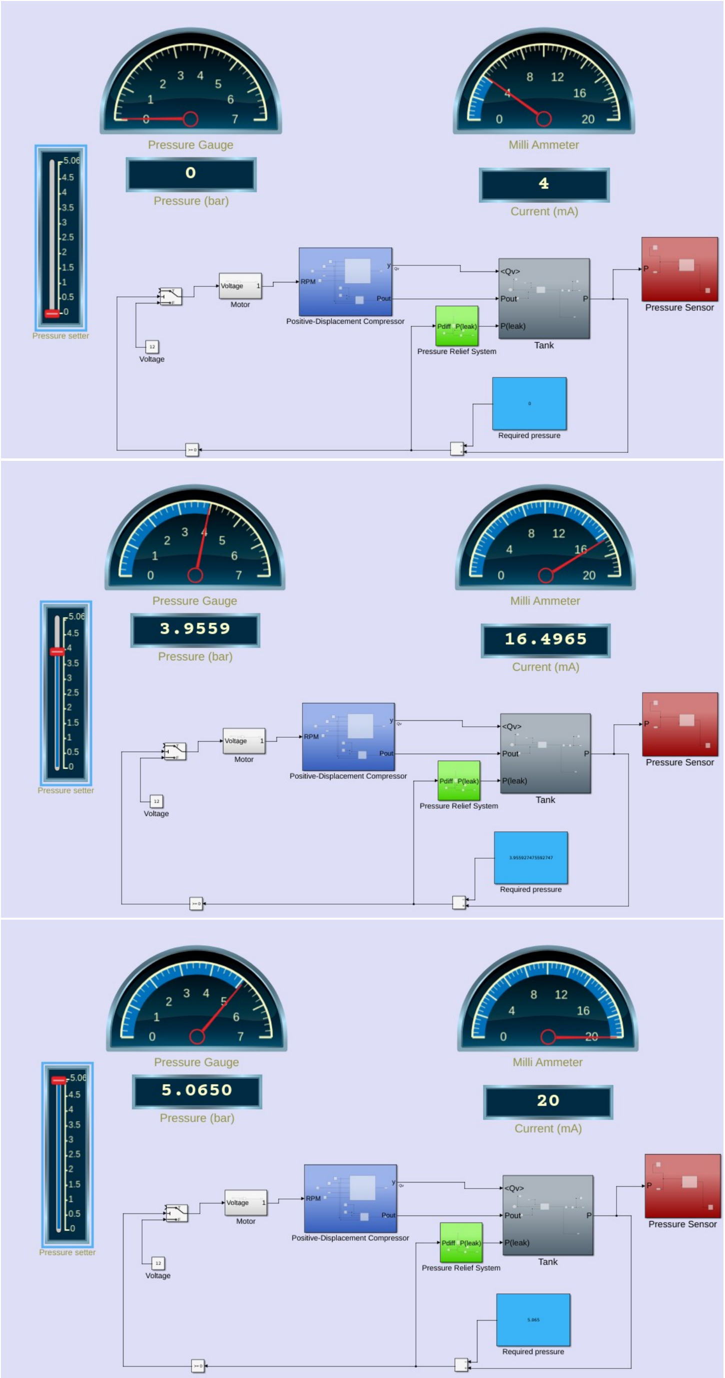

Figure 3.4: Corresponding current values (in mA) at 0, 3.955, 5.065 bar guage pressures'

CHAPTER 4

CONCLUSION AND FUTURE SCOPE

4.1 Conclusion

This project successfully developed a MATLAB Simulink–based model of a pneumatic pressure generation, regulation, and monitoring system using a simplified practical positive-displacement compressor. The system integrates electrical input control, mechanical actuation, pneumatic pressure storage, safety relief logic, and industrial-standard pressure sensing into a unified simulation framework.

The simulation results demonstrate stable pressure build-up, effective pressure regulation through switching and relief mechanisms, and accurate pressure-to-current conversion using the 4–20 mA standard. Each subsystem performs its intended function, and the overall system behavior aligns with theoretical expectations and practical pneumatic system operation.

The project highlights the effectiveness of simulation-based analysis for understanding complex multidisciplinary systems and provides a clear educational platform for studying pneumatic pressure control.

4.2 Limitations of the Present Model

The present model incorporates certain simplifications to maintain clarity and suitability for laboratory-level study. These include neglecting detailed motor electrical dynamics, assuming isothermal air behavior, and representing the pressure relief mechanism using functional logic rather than mechanical modeling. While these assumptions limit model fidelity, they do not compromise its instructional value.

4.3 Final Remarks

The work presented in this project demonstrates a structured and systematic approach to pneumatic system modeling using MATLAB Simulink. The model serves as a strong foundation for further research, experimentation, and advanced control system development in pneumatic applications.